

Robust prediction of response to immune checkpoint blockade therapy in metastatic melanoma

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Immune checkpoint blockade (ICB) therapy provides remarkable clinical gains and has been very successful in treatment of melanoma. However, only a subset of patients with advanced tumors currently benefit from ICB therapies, which at times incur considerable side effects and costs. Constructing predictors of patient response has remained a serious challenge because of the complexity of the immune response and the shortage of large cohorts of ICB-treated patients that include both ‘omics’ and response data. Here we build immuno-predictive score (IMPRES), a predictor of ICB response in melanoma which encompasses 15 pairwise transcriptomics relations between immune checkpoint genes. It is based on two key conjectures: (i) immune mechanisms underlying spontaneous regression in neuroblastoma can predict melanoma response to ICB, and (ii) key immune interactions can be captured via specific pairwise relations of the expression of immune checkpoint genes. IMPRES is validated on nine published datasets^{1–6} and on a newly generated dataset with 31 patients treated with anti-PD-1 and 10 with anti-CTLA-4, spanning 297 samples in total. It achieves an overall accuracy of AUC = 0.83, outperforming existing predictors and capturing almost all true responders while misclassifying less than half of the nonresponders. Future studies are warranted to determine the value of the approach presented here in other cancer types.

Melanoma, even in its metastatic form, is one of a handful of cancers in which spontaneous regression has been frequently observed and has been tightly linked to immune response^{7,8}. This led us to conjecture that the immune components governing spontaneous tumor regression may be a major determinant of immune responses to ICB. To this end, we focused on neuroblastoma (NB), for which we could take advantage of an existing cohort of patients with transcriptomic and clinical outcome data. Interestingly, NB in children less than 18 months of age manifests frequent spontaneous regression⁹ that is mediated by cellular immunity, including tumor-infiltrating lymphocytes, tumor-targeted T cells and antineural antibodies¹⁰. Moreover, NB is the first pediatric cancer with an Food and Drug Administration (FDA)-approved immunotherapy (dinutuximab), a monoclonal antibody targeting the disialoganglio-

side GD2 that is expressed in NB, melanoma and other tumors^{11,12}. We thus hypothesized that an immune-based predictor of NB spontaneous regression may effectively predict ICB response in melanoma.

To test this hypothesis, we built a predictor of spontaneous regression in NB, analyzing the transcriptomics data of 108 patients. Those include both spontaneously regressing (patients considered low-risk NB who have no tumor progression) and high-risk progressing patients (i.e., without spontaneous regression; Methods)¹³. We focused on 28 immune checkpoint genes collected from the literature that were included in all RNA-sequencing (RNA-seq) datasets available to us (Supplementary Table 1). We based the NB predictor on pairwise relations between the (normalized) expression levels of these genes. Each predictive feature compares the expression of two checkpoint genes A and B, capturing a logical relation between their transcriptional levels (for example, A > B). We performed a feature-selection procedure searching for a subset of these features that best separates patients with spontaneously regressing NB from those with high-risk progressing disease, resulting in the 15 most predictive features (Methods). Based on these features, the prediction of spontaneous regression of a tumor sample from its expression data is simply made by counting the number of predictive feature pairs that are fulfilled (true) in that sample given its transcriptomics data. This number, ranging from 0 to 15, denotes its IMPRES, with higher scores predicting spontaneous regression (Supplementary Table 2 and Supplementary Methods). The resulting predictor obtains an accuracy of 0.9 (in terms of the area under the receiver operator curve (AUC)) in the NB dataset (Supplementary Fig. 1 and Supplementary Methods). Reassuringly, by examining tumors derived from patients with melanoma who were not treated with ICB¹⁴, we uncovered that the IMPRES scores of patients denoted as ‘high immune response’ are considerably higher than that of other subtypes (Fig. 1a). Additionally, we found that IMPRES is significantly and positively associated with higher overall survival in these datasets¹⁴ (Fig. 1b).

We next studied nine different melanoma datasets to investigate whether there are similarities between the cellular processes mediating spontaneous tumor regression in NB and those mediating spontaneous and ICB-stimulated immune response in melanoma^{1–6}. To this end, we identified immune-related consistently differentially expressed

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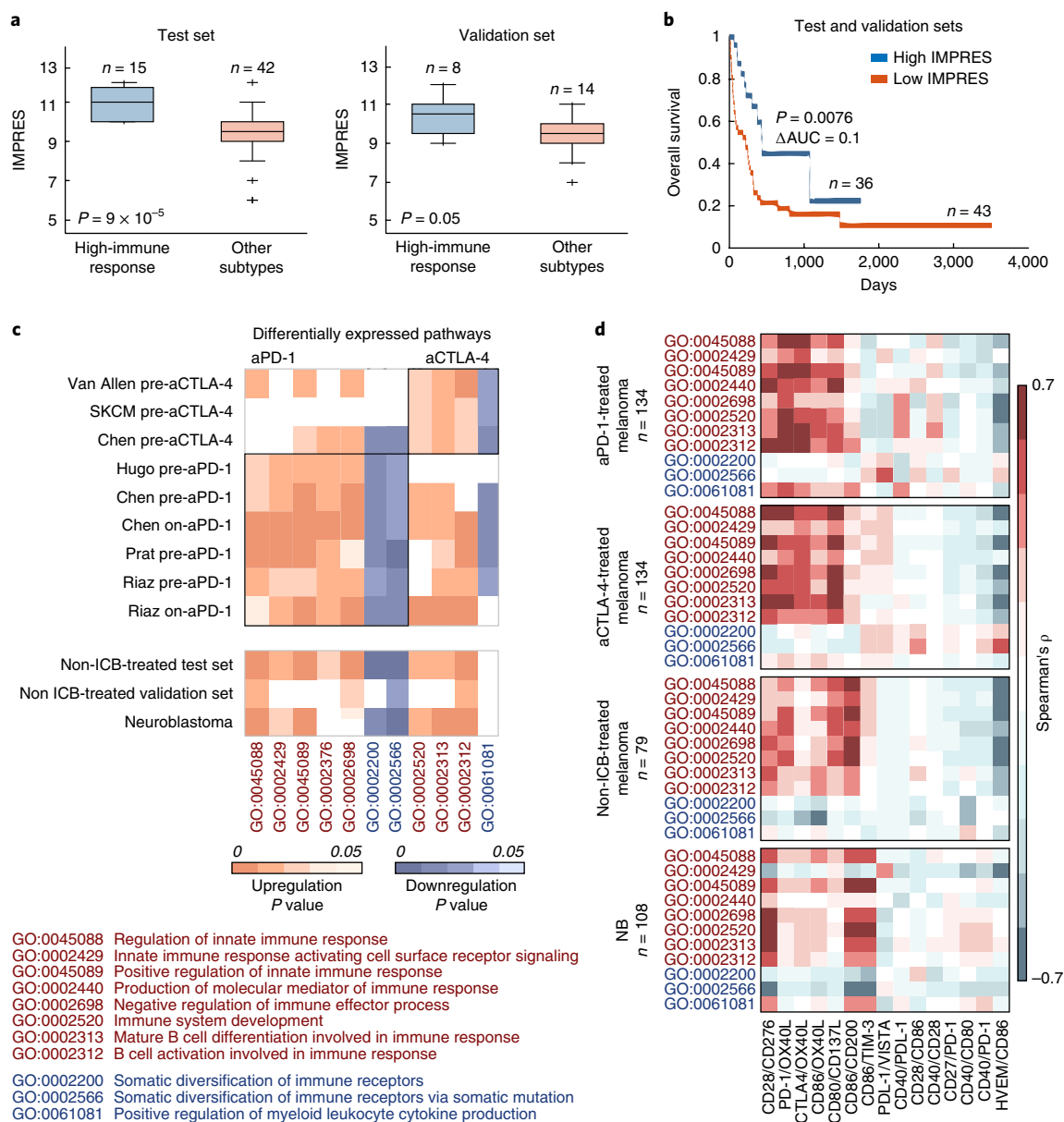


Fig. 1 | Shared immune components between melanoma and neuroblastoma. a, Boxplots showing IMPRES of high immune response versus other subtypes in test and validation datasets of non-ICB-treated patients with melanoma¹⁴. P values were computed via a one-sided rank-sum test. Boxplot center lines indicate medians, box edges represent the interquartile range, whiskers extend to the most extreme data points not considered outliers and the outliers are plotted individually using the '+' symbol. **b**, Kaplan-Meier survival curves of patients with high versus low IMPRES (computed over the combined test and validation datasets¹⁴). The median IMPRES is used to define the 'low IMPRES' and 'high IMPRES' subgroups. The P value is computed via a two-sided log-rank test. **c**, Top, Heat maps showing the hypergeometric enrichment P values for CDPs that are up- (orange) or downregulated (purple) in responders versus nonresponders across the anti-PD-1 (aPD-1; encapsulated in the left rectangle) and the anti-CTLA-4 (aCTLA-4; right rectangle) melanoma datasets (indicated on the left)^{1,3,4,6}. 'Pre', before treatment; 'on', during treatment. Bottom, enrichment P values for these CDPs in high immune response versus other subtypes in non-ICB-treated melanoma and in spontaneous regression versus nonspontaneous regression in the NB dataset; CDPs are listed with their GO identifier. **d**, Heat maps showing the Spearman rank correlation ρ between expression levels of each CDP (vertical axis) and each of the IMPRES features ratios (horizontal axis), computed separately over the anti-PD-1, anti-CTLA-4, non-ICB-treated melanoma NB datasets. White-colored entries denote associations that were not statistically significant by Spearman's rank correlation

pathways (termed CDPs) in ICB responders versus nonresponders (evaluated separately for patients treated with anti-PD-1 or anti-CTLA-4 treatments; Methods). We identified seven CDPs that were common across all anti-PD-1 datasets and four CDPs across all anti-CTLA-4 datasets. We found that these CDPs were also differentially expressed in a similar manner in the 'high immune response' melanomas compared with other subtypes¹⁴ and in spontaneously regressing versus high-risk progressing NB tumors (Fig. 1c, Supplementary Table 3

and Methods; see Supplementary Fig. 2, Supplementary Table 4 for a related analysis based on estimated immune cell abundances). We then computed the correlations between each IMPRES feature (using expression ratios; Methods) and the expression of each of the CDPs. As evident from Fig. 1d, these associations were consistently maintained across the four sample groups studied.

We next applied IMPRES to predict the response of patients with melanoma to ICB, without any further training. To this end, we

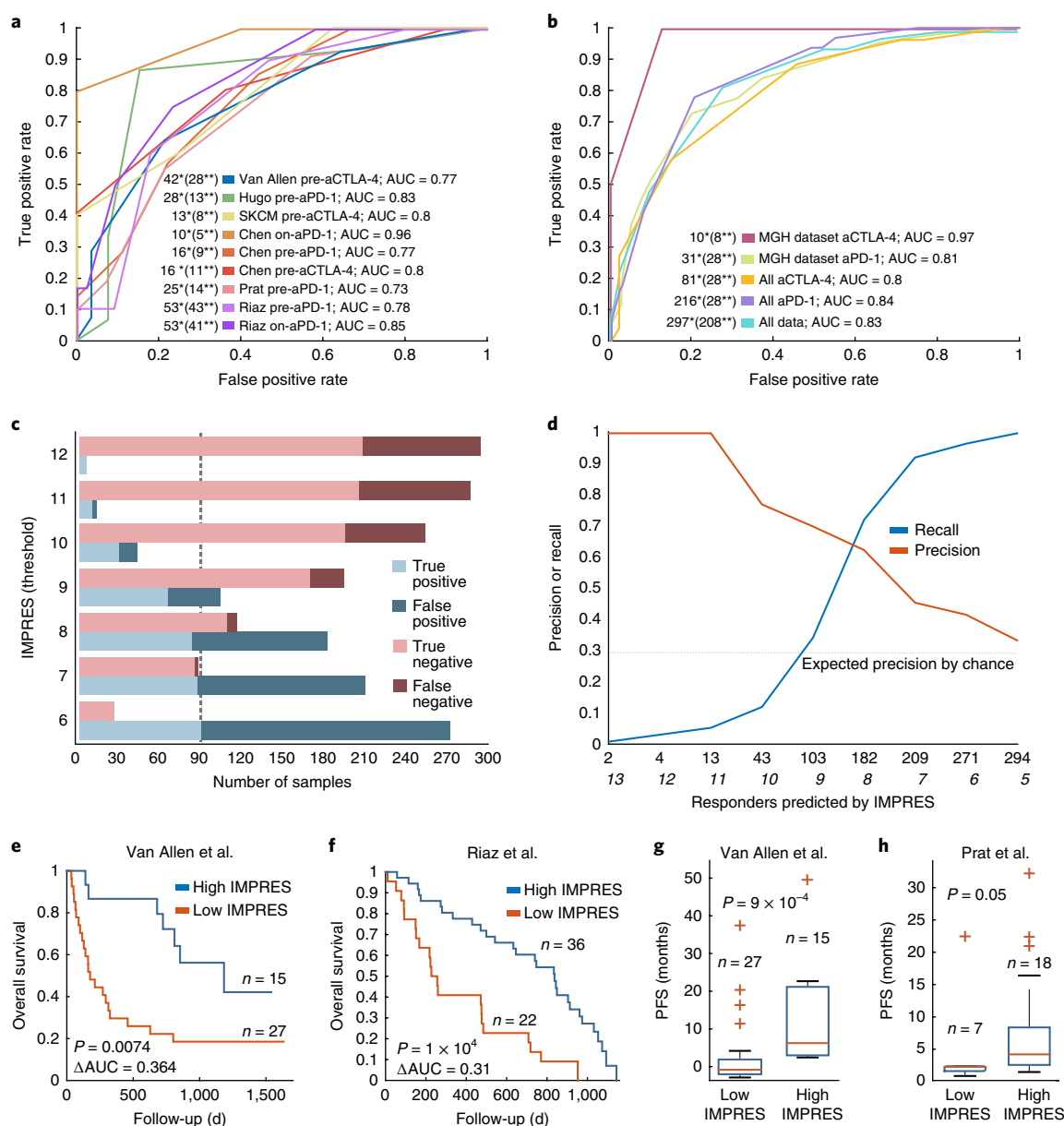


Fig. 2 | Response and survival predictive performance of IMPRES. **a**, ROC curves quantifying IMPRES prediction AUC across numerous publicly available ICB response datasets¹⁻⁶. **b**, ROC curves for the MGH dataset of ICB response (with 10 patients treated with anti-CTLA-4 and 31 patients treated with anti-PD-1) and for the aggregate datasets including all 297 samples, comprising 216 samples of patients treated with anti-PD-1 and 81 with anti-CTLA-4. In **a** and **b**, * denotes the number of samples, and ** denotes the number of nonresponding samples. **c**, Bar plots showing the prediction accuracy and error types for different IMPRES thresholds (where a positive label corresponds to a 'responder' prediction) on the aggregate compendium of the 297 patients included in all 11 datasets studied. The dashed line represents the total number of responders. **d**, Precision and recall evaluation of IMPRES on the same aggregate compendium. The y axis displays the precision or recall as a function of the number of 'responder' predictions made (shown on the x axis, obtained by decreasing the classification threshold, whose value is also displayed in italic font under the number of responders). Prediction performance in terms of specificity and sensitivity values is provided in Supplementary Table 5. **e, f**, Kaplan-Meier survival curves for the ICB treatment datasets (Van Allen et al.¹ and Riaz et al.⁶, respectively), with high versus low IMPRES scores (using the median IMPRES as a threshold differentiating between the high and low groups). The *P* values were computed via a two-sided log-rank test. **g, h**, Boxplots comparing PFS between low versus high IMPRES in the ICB (Van Allen et al.¹ and Prat et al.⁵ datasets, respectively), using the median IMPRES as a differentiating threshold. *P* values were computed via a one-sided rank-sum test. Boxplot center lines indicate medians, box edges represent the interquartile range, whiskers extend to the most extreme data points not considered outliers and the outliers are plotted individually using the '+' symbol.

analyzed 256 samples from nine datasets derived from six independent studies, including patients treated with anti-CTLA-4, anti-PD1 or their combination¹⁻⁶. We computed the IMPRES score of each melanoma sample from its expression data and used those and the clinical response data to generate the receiver operator characteristic (ROC) classification curves quantifying IMPRES prediction

performance in each of the different datasets. The resulting AUCs are in the range of 0.77–0.96 (Fig. 2a).

We further tested the predictive ability of IMPRES in a newly generated RNA-seq dataset of tumor biopsies from metastatic melanoma patients treated with ICB therapies at the Massachusetts General Hospital. IMPRES achieves AUCs of 0.81 and 0.97 on the

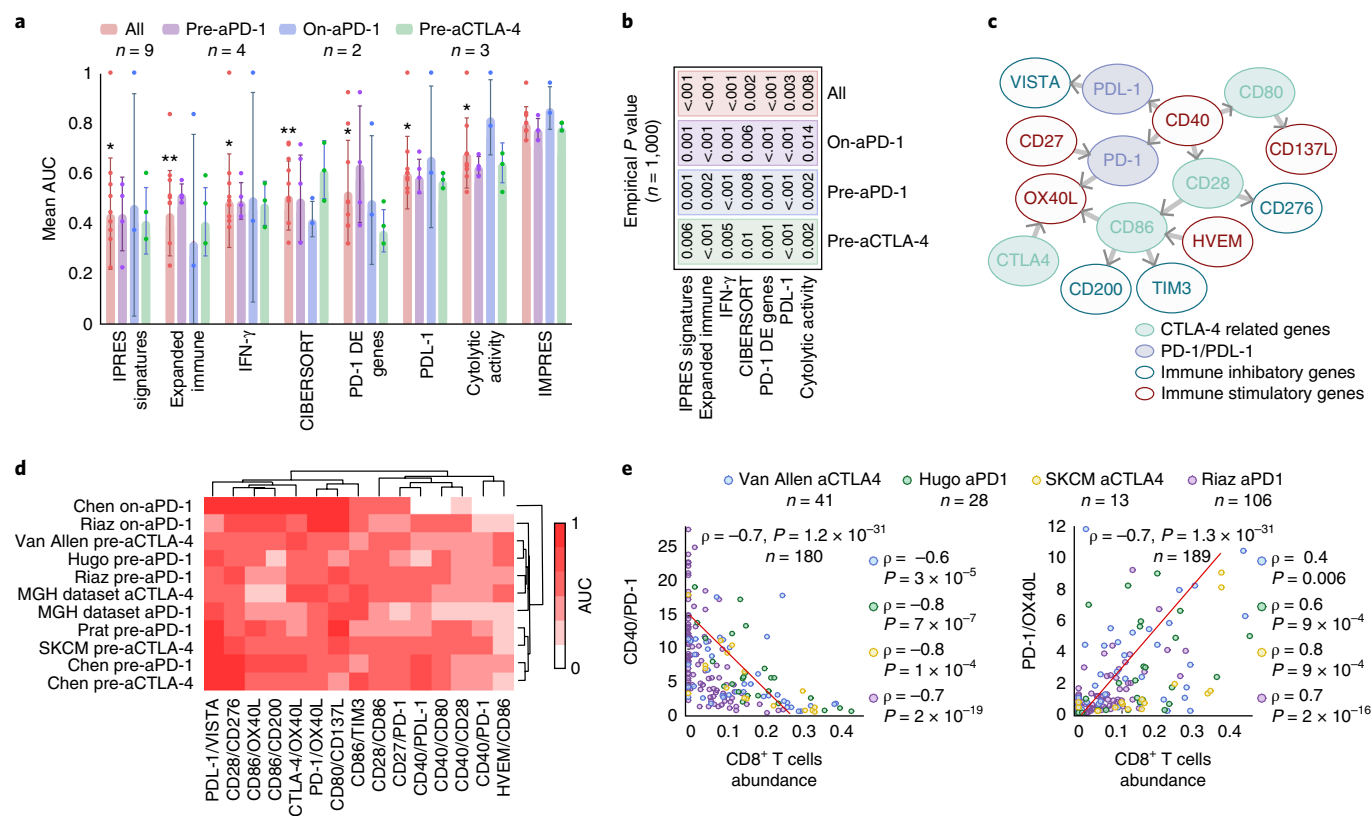


Fig. 3 | Comparative performance of IMPRES and its underlying features. a, AUC of IMPRES and other published predictors across nine publicly available ICB treatment datasets grouped by treatment type and stage. The one-sided rank-sum P values comparing the performance of each predictor evaluated to that of IMPRES over all datasets are presented (P value of 0.002 is achieved when IMPRES AUC is larger than that obtained by the other predictor for all nine datasets and of 0.004 when it is larger for eight of nine datasets). * $P=0.004$; ** $P=0.002$. The bar centers show the means, and error bars show s.d. DE, differentially expressed. **b**, The empirical P values comparing IMPRES performance to that of each of the other predictors in the three different ICB treatment classes and for the aggregate of all datasets (using $n=1,000$ permutations; the value $<.001$ denotes that the prediction performance of IMPRES was superior to that of the predictor with which it was compared in all 1,000 repetitions). **c**, A network representation of the 15 pairwise features comprising IMPRES. Each node represents an immune checkpoint gene, and each edge describes a pairwise relation (an IMPRES feature). The direction of edge $A \rightarrow B$ denotes that expression of A higher than that of B is associated with better response in patients. The outline color of each node denotes whether it is inhibitory or activating, and its fill color denotes whether it belongs to the PD-1 or CTLA-4 pathway. **d**, Clustrogram (with average linkage function) of the individual predictive power of the 15 IMPRES features (based on their expression ratios) in each of the melanoma treatment datasets studied (the color scaling denotes the AUC obtained using each individual ratio as a response predictor, ranging from 0 to 1). **e**, Scatter plots showing the correlation between CIBERSORT-inferred CD8⁺ T cells abundance (x axis) and the gene expression ratios of two IMPRES features that are significantly associated with it (y axis); CD40/PD-1 (left) and PD-1/OX40L (right). The Spearman ρ and associated P values are shown for each ICB response dataset^{1,3,4,6} individually (in the key) and for all four datasets together (in the graph).

anti-PD-1 and anti-CTLA-4 samples, respectively (Fig. 2b). It maintains its predictive accuracy when evaluating the aggregate collection of the datasets studied above (a total of 297 samples; Fig. 2b). Figure 2c shows the number of true and false positives (responders) and true and false negatives (nonresponders) obtained on this aggregated data at different IMPRES score classification thresholds, manifesting the well-known tradeoff between precision and recall (Fig. 2d, Supplementary Fig. 3a,b and Supplementary Table 5). As evident, IMPRES can capture almost all true responders while misclassifying less than half of the nonresponders (at threshold=8). Higher IMPRES scores are also associated with improved overall survival and progression-free survival (PFS) in ICB-treated patients with melanoma (Fig. 2e–h, Methods and Supplementary Fig. 3c).

To compare the predictive accuracy of IMPRES with that of current transcriptome-based predictors, we generated predictors of response to ICB for each published transcriptomic signature (Methods). The performance of IMPRES was superior to the other predictors (Fig. 3a and Supplementary Table 6). This observation also held true when we compared the performances on each ICB

treatment group separately (Fig. 3b). Overall, the predictors built on biologically motivated scores (cytolytic activity¹⁵ and PDL-1 expression) generalized better than the machine learning–based predictors constructed on transcriptomic signatures identified in the specific cohorts. Of note, although we found a significant correlation between IMPRES and CD8⁺ and CD4⁺ T cells abundances inferred via CIBERSORT, the latter are poor predictors of ICB response (Supplementary Fig. 4). IMPRES superiority is particularly notable because for most existing signature-based predictors (all but cytolytic-activity¹⁵ and PDL-1 expression), we had to retrain the latter separately for each dataset; otherwise, their overall performance was dismal, testifying to their poor generalizability between different datasets (Methods). In contrast, IMPRES is constructed only once from the NB data and never trained on any melanoma dataset. Thus, it is markedly less prone to overfitting, a paramount concern regarding standard cancer transcriptomics predictors^{16–18}. To further study the importance of training on the independent NB data, we trained ICB response predictors based on melanoma data instead of NB, following exactly the same representation and

training procedure as used in IMPRES. This resulted in markedly lower prediction performances on the melanoma datasets that were not used for training compared to the original IMPRES procedure (Supplementary Fig. 5). Finally, IMPRES performance remained superior when this comparative analysis was repeated while excluding patients annotated with 'stable disease' (Supplementary Table 6).

The features comprising IMPRES lead to a few insights that are biologically interesting. Reassuringly, the relatively higher expression of genes encoding immune stimulatory molecules (such as HVEM, CD27 and CD40) is associated with a better response to ICB, whereas the higher expression of genes encoding immune inhibitory molecules (such as CD276, TIM-3, CD200 and VISTA) is associated with a worse response, as expected (Fig. 3c). Higher expression of CD40 compared to that of PD-1, PDL-1, CD80 and CD28 is associated with a better ICB response, in line with the recent finding that agonists of CD40 reverse resistance to anti-PD-1 therapy and that induced PD-1 expression mediates acquired resistance to antagonist CD40 treatment¹⁹. Additionally, the higher expression of CD27 compared to that of PD1 (but not compared to CTLA-4) is associated with improved response. This is in line with recent findings that the combination of a CD27 agonist plus anti-PD-1 recapitulates the effects of CD4⁺ T helper cells on tumor control, whereas the combination of a CD27 agonist plus anti-CTLA-4 did not improve tumor control²⁰.

We further studied the individual predictive power of the IMPRES features by considering the expression ratio of each predictive pair (Methods). We find that some features are specifically more predictive for anti-PD-1 pretreatment (CD28/CD86, rank-sum $P=0.05$) or on-treatment (PD-1/OX40L, CD86/OX40L and CD86/CD200, rank-sum $P=0.018$ for all; Supplementary Table 7c). Notably, no feature emerges as being strongly predictive of response to anti-CTLA-4 specifically (Fig. 3d and Supplementary Table 7c). Examining the associations between these 15 features (using their expression ratio) and the inferred abundance of 22 types of immune cells (Supplementary Table 4) uncovers two significant associations, CD40/PD-1 and PD1/OX40L (Fig. 3e). Finally, a feature reduction analysis (Methods) shows that the overall predictive performance of IMPRES can be maintained with a subset of 11 of the 15 original features, but beyond that it markedly decreases (Supplementary Fig. 6 and Supplementary Table 8).

In summary, the high predictive performance of IMPRES is mainly due to two key conjectures: (i) key immune mechanisms underlining spontaneous regression in NB are shared with those determining response to ICB in melanoma, and (ii) those may be captured by specific pairwise relations of expression of immune checkpoint genes. Building on these assumptions leads to a predictor of response to checkpoint therapy that is substantially superior to existing predictors and displays robust performance across many different melanoma datasets. From a translational standpoint, we show that IMPRES can correctly capture almost all true responders while misclassifying less than half of the nonresponders. Future studies are warranted to further study the predictive performance of the approach presented here in other cancer types for which ICB is approved as sufficiently large datasets are accumulated.

Methods

Methods, including statements of data availability and any associated accession codes and references, are available at <https://doi.org/10.1038/s41591-018-0157-9>.

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Author contributions

E.R. supervised the research. N.A. and E.R. conceived and designed the computational approach. N.A. performed statistical and machine-learning analysis. N.A. and J.S.L. analyzed the data. S.M. applied the CIBERSORT deconvolution tool. E.R., N.A. and G.Z. wrote the paper. M.H., K.F., G.B. and J.S.L. helped write the paper. G.Z., B.M., D.T.F., T.M., T.T., Z.W. and R.J.S. collected the clinical and transcriptomic data under the supervision of G.Z., K.F., G.B. and M.H.

Competing interests

The authors declare no competing interests.

Additional information

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Methods

Ethics statement. All clinical data and patient samples were collected following approval by the Massachusetts General Hospital Institutional Review Board (IRB). In all cases, informed consent was obtained from patients.

Collection of immune checkpoint molecules. To build a predictor based on pair-wise relations between expression of checkpoint genes, we formed a list of 45 immune checkpoint genes with known costimulatory or co-inhibitory effects, collected from literature reports^{21–24}. From these, we focused on 28 genes that were measured in all RNA-seq datasets analyzed in this paper (Supplementary Table 1).

Feature selection and IMPRES construction on the NB data. For feature selection, we use the quantile-normalized expression of the 28 immune checkpoint genes selected above in the 108 NB tumor samples studied (Supplementary Methods), using the following expression function of pairs of checkpoint genes as features:

$$F_{ij}(x) = \begin{cases} 1, & \exp_i(x) < \exp_j(x) \\ 0, & \text{otherwise,} \end{cases}$$

Where $\exp_i(x)$ and $\exp_j(x)$ denote the expression of genes i and j in sample x .

We focus on pairs where at least one of the genes is among the six genes that are directly associated with anti-CTLA-4 and anti-PD1 blockade therapy, including the genes encoding CTLA-4, CD28, CD80/CD86, PD-1 and PD-L1 (ref. 25), which together form 294 potential gene pairs. To select features that best separate positive from negative samples in the NB data, we performed a hill-climbing aggregative feature selection involving 500 iterations of a fivefold cross-validation procedure, in which the features that highly scored consistently across folds were selected for IMPRES. A detailed description of the feature selection steps is available in the Supplementary Methods.

Immune pathway enrichment analysis. To identify CDPs, we first identified the genes that are up- and downregulated in ICB responders versus nonresponders for each of the datasets^{1,3,4,6} (using one sided rank-sum $P < 0.05$). Then, we performed a gene ontology (GO) pathway²⁶ enrichment analysis for immune-related pathways (Supplementary Table 3) via a hyper-geometric test, to identify (i) pathways that are consistently up- or downregulated (hypergeometric $P < 0.05$) in responders for all anti-PD-1 melanoma datasets, and (ii) pathways that are consistently up- or downregulated in responders for all anti-CTLA-4 melanoma datasets (Fig. 1c and Supplementary Table 3).

To correlate CDPs with the IMPRES features, we then evaluated the Spearman rank correlation coefficients (ρ) and corresponding P values between the median pathway expression level of each CDP (using the median expression of all genes in a pathway) and each of the IMPRES expression ratios. This was done across all samples in each of the following datasets: (i) the anti-PD-1-treated melanoma datasets, (ii) the anti-CTLA-4-treated melanoma datasets, (iii) the non-ICB-treated melanoma datasets and (iv) the NB dataset.

Computing expression ratios of IMPRES features. To evaluate the predictive performance and functional associations of individual IMPRES features in a more refined manner, we used the expression ratio instead of the binary indicators in each sample (i.e., for each feature $A > B$, we used A / B instead). The resulting AUCs obtained with each ratio feature for each ICB response data are presented in Fig. 3d and Supplementary Table 7b.

CIBERSORT analysis. Using CIBERSORT²⁷ we infer the relative abundances of 22 immune cells in NB and melanoma samples (analyzing each dataset for ICB-treated and nontreated melanoma). Having estimated these cell abundances for each sample, we perform the following analyses:

1. We perform a differential abundance analysis via a one-sided rank-sum test for each immune cell type between NB samples with or without spontaneous regression, between melanoma samples from responders and nonresponders to ICB treatment, and between high immune response versus other subtypes in non-ICB-treated melanoma samples (Supplementary Table 4).
2. To study the relation between IMPRES and major T cell types mediating the immune response, we correlated the CIBERSORT inferred relative abundances of CD8⁺ and CD4⁺ T cells to IMPRES scores for RNA-seq melanoma ICB response datasets via Spearman's rank correlation (Supplementary Fig. 4).
3. To survey the key associations between IMPRES features and immune subtypes, we correlated each CIBERSORT inferred immune cell type to each IMPRES feature (considering expression ratios instead of binary relations), for RNA-seq melanoma ICB response datasets via Spearman's rank correlation (Supplementary Table 4).

Applying IMPRES to predict ICB response of patients with melanoma. To apply IMPRES, we calculate for each sample x , the $F_{ij}(x)$ over the 15 IMPRES checkpoint pairs (features). This leads to a binary vector of length 15 for each sample. The total

number of '1's in this vector denotes the sample's IMPRES score (ranging between 0 and 15). High scores predict good response. By varying the classification threshold over the different possible IMPRES score values we generate the ROC curves and the resulting AUCs presented in the main text for each melanoma dataset.

RNA-seq. RNA-sequencing of 31 anti-PD-1 pre- and on-treatment tumor specimens and 10 anti-CTLA-4 pre- and on-treatment metastatic tumor specimens (for which the response is known) derived from patients with metastatic melanoma (up to 90 days from treatment start) was conducted as previously described in Jenkins et al.²⁸ (Supplementary Table 9). These patients were enrolled in clinical trials at Massachusetts General Hospital. Clinical trial registration numbers at ClinicalTrials.gov are NCT01714739; NCT02083484; NCT01543698; NCT01072175; NCT00949702; NCT01783938; and NCT01006980.

Clinical response classification. Supplementary Table 10 summarizes the response annotations and criteria used for establishing them in the original study. The response classification of each patient in each of the publicly available studies and the MGH dataset (with MGH patients clinical information) is described in Supplementary Table 9.

To apply IMPRES and evaluate its predictive performance on the combination of melanoma samples in all 11 datasets studied, we normalize the IMPRES scores for datasets in which not all relevant checkpoint genes were measured (Supplementary Table 1). This normalization is done by linearly scaling the IMPRES score to compensate for the number of missing pairs whose expression is not available in the dataset. For example, if 13 out of the 15 IMPRES features are measured in a dataset, then the IMPRES score of these samples is multiplied by 15 and divided by 13 to linearly scale it back to the original scale of 0–15. We then calculate the AUC in a standard manner using these normalized IMPRES scores over all samples.

Training a predictor using melanoma datasets. We use a similar training procedure as described above for IMPRES (using binary relations ($A > B$) between the 28 checkpoint genes as before), but this time training on melanoma datasets: (i) first, we train on the combined data from Riaz et al.⁶ and Hugo et al.⁹ (both anti-PD-1 datasets), and (ii) we then train on the combined datasets from Hugo et al. and Van Allen et al.^{1,3} (anti-PD-1 and anti-CTLA-4 datasets, respectively). For both, we use hill climbing feature selection and perform similar procedures as described above in 'Feature selection and IMPRES construction on the NB data', of 500 rounds of fivefold training and testing. The final feature set is also selected in the same manner (using similar definition of $score(f)$ and selecting features with binomial $P < 0.05$).

IMPRES analysis of different melanoma subtypes. We evaluated whether IMPRES significantly differed between different melanoma subtypes by comparing the IMPRES score of each subtype against that of all other subtypes when using pre-anti-PD-1, on-anti-PD-1 and all samples from Riaz et al.⁶ (Supplementary Figure 7).

Comparing IMPRES predictive performance to that obtained by predictors based on other published signatures. We compared the performance of IMPRES to those obtained using other published transcriptomic signatures^{3,15,29} as well as PDL-1 expression. We additionally evaluated the performance of a predictor based on immune cell abundances estimated via CIBERSORT²⁷. The predictors' performance is evaluated using the nine publicly available melanoma datasets analyzed to evaluate IMPRES (see main text). The Cytolytic activity¹⁵ and PDL1 expression-based predictors are applied in a straightforward manner, analogous to that of IMPRES, as they do not require additional training. However, making predictions using gene signatures reported in specific studies in the literature (see main text) requires training on every specific dataset tested (using cross-validation), aiming to identify their maximal performance levels. Hence, we build predictors of ICB response using Support Vector Machines (SVMs) on each of the pertaining melanoma datasets. Each such SVM predictor is built using the genes in the specific signatures on which it is based as its feature set. This is performed with linear kernels using 100 repetitions of a fivefold cross-validation process, wherein each fold the training set and test set are randomly selected. The AUC presented for each predictor is the mean AUC overall repetitions (Supplementary Table 6).

To compare IMPRES performance to that of other predictors over different treatment groups in a systematic manner, we aggregate the samples into four treatment groups: pre-anti-CTLA-4, pre-anti-PD-1, on-anti-PD-1 and all samples. To calculate an empirical P value, we then perform 1,000 repetitions of: (i) randomly sampling 80% of the samples in a stratified manner (maintaining the proportion of responders versus nonresponders) from each treatment group and (ii) evaluating the AUC resulting from each predictor on the randomly selected samples from each treatment group. The resulting empirical P value (reported in Fig. 3b) denotes the percentage of iterations in which the AUC obtained via each predictor is superior to that obtained using IMPRES for each treatment group.

Feature reduction analysis. As IMPRES features were selected for NB, it is possible that some are less predictive for melanoma response. To investigate which

features may be removed, we performed a feature-removal procedure using fivefold cross-validation on the combined data from all five RNA-seq datasets^{1,3,4,6} (as these cover all IMPRES features). For each fold, we performed 500 rounds of greedy (hill-climbing) feature removal on the training set. Each round ended when the performance (AUC) was reduced by more than 5% from the full set performance on the training set. The set of remaining features was applied to the test set to obtain a test AUC. We found a group of 11 features that captured most of IMPRES original performance levels (Supplementary Fig. 6 and Supplementary Table 8). All groups with less than 11 features have reduced the test AUC by more than 5%.

IMPRES sensitivity analysis and random control predictors. To evaluate IMPRES sensitivity to missing features, we performed all possible removals of single, double, triple and quadruple features. For each removal, we examine the AUCs obtained with the remaining features for each of the 11 datasets. We found that, although the results remained robust for most single- and double-feature removals, higher-order removals are likely to reduce the performance, but not in all cases (for example, as in the reduced features set; Supplementary Fig. 6 and Supplementary Table 11).

To evaluate the power of predictors constructed via randomly selected relations, we generated 1,000 predictors, each based on randomly selected 15 immune gene relations (drawing from the same pool of features as those considered for the construction of IMPRES, Supplementary Fig. 8).

Principle component analysis using IMPRES features. We perform principle component analysis (PCA) of patient profiles for each melanoma ICB study and across all studies combined using IMPRES selected features (i.e., each sample is represented as a 15-dimensional binary vector comprised of the 15 IMPRES logical relations). The PCA results are presented in Supplementary Fig. 9.

Comparing IMPRES scores to mutational counts across TCGA tumors.

Examining pan-cancer TCGA data, we find that cancer types with high IMPRES scores tend to have a higher mutational burden, a well-established marker of response to immunotherapy (Spearman's $\rho = 0.79$; Supplementary Fig. 10). To perform this analysis, the complete raw data of all TCGA cancer types ($n = 23$) which include at least 100 patients were downloaded from cBioPortal³⁰. We considered samples containing somatic point mutations and gene expression data, relative to matched-normal samples.

Statistical analysis. Boxplots and comparisons. For all boxplots, center lines indicate medians, box edges represent the interquartile range, whiskers extend to the most extreme data points not considered outliers, and the outliers are plotted individually using the '+' symbol. Points are defined as outliers if they are greater than $q_3 + w \times (q_3 - q_1)$ or less than $q_1 - w \times (q_3 - q_1)$, where w is the maximum whisker length, and q_1 and q_3 are the 25th and 75th percentiles of the sample data, respectively. All differential expression and distribution comparison P values are obtained via one-sided rank-sum test.

Survival analyses. All Kaplan–Meier analyses were performed by comparing the survival of patients with high scores ($>$ median) to those with low scores ($<$ median) using a two-sided log-rank test. The patients with median score ($=$ median) were grouped with the smaller-size group among the two groups mentioned above.

Bar plots. For bar plots, the center is defined by the mean of the distribution, and error bars represent the s.d. of the distribution.

Correlation coefficients. All correlations coefficients and P values were obtained via Spearman's rank correlation test.

Code availability. Codes are implemented in MATLAB and are publicly available in GitHub: <https://github.com/noamaus/IMPRES-codes>.

The CIBERSORT software was applied on matrices of gene expression data using the given LM22 immune cell signatures.

Reporting Summary. Further information on experimental design is available in the Nature Research Reporting Summary linked to this article.

Data availability. All patients data analyzed from published papers are referenced to and publicly available accordingly. The transcriptomic data of the MGH patients analyzed are available from GEO (GSE115821). MGH patients' clinical information is available in Supplementary Table 9.

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► Experimental design

1. Sample size

Describe how sample size was determined.

Only samples that were available for this study and passed quality control were included in the sequencing queue

2. Data exclusions

Describe any data exclusions.

All samples for which the response was known at time of submission were included

3. Replication

Describe whether the experimental findings were reliably reproduced.

Experimental replications were not attempted

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

The cohort of patients were already randomized as they were participants in clinical trials

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

The investigators were blinded to the response annotations until they became available

Note: all studies involving animals and/or human research participants must disclose whether blinding and randomization were used.

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.)
- ☐ ☒ A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ A statement indicating how many times each experiment was replicated
- ☐ ☒ The statistical test(s) used and whether they are one- or two-sided (note: only common tests should be described solely by name; more complex techniques should be described in the Methods section)
- ☐ ☒ A description of any assumptions or corrections, such as an adjustment for multiple comparisons
- ☐ ☒ The test results (e.g. P values) given as exact values whenever possible and with confidence intervals noted
- ☐ ☒ A clear description of statistics including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range)
- ☐ ☒ Clearly defined error bars

See the web collection on [statistics for biologists](#) for further resources and guidance.

► Software

Policy information about [availability of computer code](#)

7. Software

Describe the software used to analyze the data in this study.

MATLAB R2014B, R2018A and R2017B, CIBERSORT

For manuscripts utilizing custom algorithms or software that are central to the paper but not yet described in the published literature, software must be made available to editors and reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). *Nature Methods* [guidance for providing algorithms and software for publication](#) provides further information on this topic.

► Materials and reagents

Policy information about [availability of materials](#)

8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a for-profit company.

No unique materials were used

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).

No antibodies were used

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.

No eukaryotic cell lines were used

b. Describe the method of cell line authentication used.

No eukaryotic cell lines were used

c. Report whether the cell lines were tested for mycoplasma contamination.

No eukaryotic cell lines were used

d. If any of the cell lines used are listed in the database of commonly misidentified cell lines maintained by [ICLAC](#), provide a scientific rationale for their use.

No eukaryotic cell lines were used

► Animals and human research participants

Policy information about [studies involving animals](#); when reporting animal research, follow the [ARRIVE guidelines](#)

11. Description of research animals

Provide details on animals and/or animal-derived materials used in the study.

No animals were used in the study

Policy information about [studies involving human research participants](#)

12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.

31 tumor biopsies from patients treated with anti-PD1 and 10 from patients treated with anti-CTLA4, patients age ranges from 28 to 84 (average age of 64), where the gender of 54% of the samples is Male (M).